

BRAC University

Department of Computer Science and Engineering

3D Car Racing Arena

Project Report

Course: CSE423

Course Title: Computer Graphics

1. Project Overview

For our CSE423 Computer Graphics project, we developed a 3D car racing game titled **3D Car Racing Arena**. The main idea of the project was to create an interactive racing environment where the player can control a car, move around a 3D track, avoid obstacles, collect coins, shoot enemies, and complete the game by surviving different challenges.

We built the project using Python and OpenGL. Instead of using a game engine, we created the objects and game environment using basic graphics functions, shapes, transformations, and camera control. This helped us apply the computer graphics concepts that we learned in the course in a practical way.

The project includes a racing arena, player car, enemy cars, coins, traps, obstacles, bullets, weather effects, health system, score system, boost option, cheat modes, camera switching, and win/loss conditions. Overall, the project works as a complete 3D racing and survival game.

2. What We Built

In this project, we built a 3D racing arena where the player controls the main car. The player can move the car forward, backward, left, and right. The car can also rotate and shoot bullets. The main goal of the game is to survive inside the arena, collect coins, avoid traps and obstacles, fight enemies, and reach the winning condition.

The game environment contains a road and surrounding arena objects. We added enemy cars to make the game more challenging. These enemies act as obstacles and threats for the player. The player can shoot bullets to attack them. Coins are placed in the arena so that the player can collect them and increase the score.

We also added traps and obstacles in different positions. If the player hits these objects, the health can decrease. This makes the game more difficult and requires the player to drive carefully. The game also has weather effects such as rain and fog to make the environment look more dynamic.

3. Tools and Technologies Used

The project was developed using the following tools and technologies:

Tool / Technology	Purpose
Python	Used as the main programming language.
OpenGL	Used to create and render the 3D graphics scene.
PyOpenGL	Used to access OpenGL functions in Python.
GLUT	Used for window creation, keyboard input, mouse input, and display functions.
GLU	Used for camera and perspective setup.
Math Functions	Used for movement, rotation, bullet direction, and collision checking.

Table 1: Tools and technologies used in the project

4. Main Features

4.1. 3D Racing Arena

The main environment of the project is a 3D racing arena. The arena contains the road, boundaries, and other objects required for the racing game. The player drives the car inside this arena. The environment was created using basic OpenGL drawing methods and transformations.

4.2. Player Car

The player car is the main object controlled by the user. We created the car using 3D shapes and placed it inside the arena. The player can move the car in different directions and rotate it. The movement of the car is controlled using keyboard input.

4.3. Enemy Cars

Enemy cars were added to make the game more challenging. These cars create danger for the player. The player has to avoid them or shoot them. This gives the game both racing and action elements.

4.4. Shooting System

We added a bullet shooting system to the player car. When the player shoots, bullets move forward based on the direction of the car. If a bullet hits an enemy, the enemy can be destroyed. This feature makes the game more interactive.

4.5. Coins

Coins are placed inside the arena. The player can collect coins by driving the car through them. When coins are collected, the score increases. This gives the player another objective besides only driving.

4.6. Obstacles and Traps

We added obstacles and traps in the arena. These objects create difficulty for the player. If the player hits them, the health may decrease. The player needs to avoid these objects while driving.

4.7. Health and Score System

The game has a health system and a score system. The health decreases when the player hits dangerous objects or enemies. The score increases when the player collects coins or defeats enemies. These systems help track the player's progress.

4.8. Weather Effects

The game includes weather effects such as rain and fog. These effects make the scene look more interesting and dynamic. They also show how visual effects can be added in a computer graphics project.

4.9. Camera Views

We added multiple camera views in the project. The player can switch between different views, such as a third-person view and another camera perspective. This helps the player see the arena from different angles and also demonstrates camera transformation.

4.10. Boost and Cheat Modes

The game includes a boost feature that allows the player to move faster for a certain time. We also added cheat modes for testing and gameplay support. These features help control the game state and make the project more flexible.

4.11. HUD and Buttons

The project includes a HUD, which shows important game information such as score, health, and game status. We also added buttons for game interaction. This makes the game easier to understand and use.

5. Computer Graphics Concepts Used

5.1. 3D Object Drawing

We used basic 3D shapes to create the cars, coins, obstacles, traps, road, and other objects. By combining simple shapes, we were able to create a complete game environment.

5.2. Translation

Translation was used to move objects from one position to another. The player car, bullets, enemies, and other moving objects use translation. When the player presses movement keys, the position of the car changes.

5.3. Rotation

Rotation was used to turn the car left and right. It was also used to control the direction of the car and bullets. Rotation is important because the car needs to move according to the direction it is facing.

5.4. Scaling

Scaling was used to change the size of objects. Different objects such as cars, coins, traps, and obstacles needed different sizes, so scaling was used to adjust them properly.

5.5. Camera Control

Camera control was used to change how the player sees the game. We used camera positioning and viewing transformations to create different camera modes. This helped make the game more playable and visually better.

5.6. Animation

Animation was created by continuously updating object positions and redrawing the scene. The car movement, bullet movement, enemy behavior, and weather effects are all examples of animation in this project.

5.7. Collision Detection

Collision detection was one of the most important parts of the project. We used collision checking to detect when the player car touches coins, enemies, traps, or obstacles. We also used it to detect when bullets hit enemy cars.

6. Game Logic

The game runs using a continuous loop. In each frame, the program checks user input, updates the position of objects, checks collisions, updates the score and health, and redraws the scene.

The basic working flow of the game is:

1. Start the game and initialize the OpenGL window.
2. Draw the racing arena and all objects.
3. Take keyboard and mouse input from the user.
4. Move the player car based on the input.
5. Update bullets, enemy cars, coins, traps, and weather effects.
6. Check collisions between the player and other objects.
7. Update health, score, and game status.
8. Check whether the player has won or lost.
9. Redraw the scene continuously.

7. Challenges Faced

One of the main challenges was controlling the movement of the car in a 3D space. The car needed to move according to its rotation angle, so we had to handle both position and direction properly.

Another challenge was collision detection. Since the game contains many objects such as enemies, coins, bullets, traps, and obstacles, the program had to check collisions continuously. If the collision logic was not handled properly, the gameplay would not work correctly.

Camera control was also challenging. The camera needed to follow the player and also provide different views. We had to adjust the camera position carefully so that the game would be easy to play.

Weather effects and HUD design also required extra work. These features had to be added without disturbing the main 3D scene.

8. Limitations

Although the project works as a complete 3D racing game, there are still some limitations. The object models are made using basic OpenGL shapes, so they are not as detailed as modern game models. The enemy movement is also simple and can be improved with better AI.

The collision detection is basic and does not use an advanced physics engine. The game also does not include realistic car physics, sound effects, or advanced textures. These features can be added in the future to improve the project.

9. Future Improvements

In the future, the project can be improved by adding better car models, textures, sound effects, and background music. The enemy cars can be made smarter using improved AI. More tracks and levels can also be added to make the game longer and more interesting.

We can also add a better physics system so that the car movement feels more realistic. A start menu, pause menu, and difficulty levels can also improve the user experience. Multiplayer mode can also be added as a future feature.

10. Conclusion

In this project, we created a 3D car racing arena using Python and OpenGL. The project includes player movement, enemy cars, bullets, coins, traps, obstacles, weather effects, camera views, health, score, and game-ending conditions.

Through this project, we were able to apply many important computer graphics concepts such as 3D object drawing, translation, rotation, scaling, animation, camera transformation, and collision detection. The project helped us understand how graphics concepts are used together to build an interactive 3D application.

Overall, the project was a practical implementation of the topics learned in CSE423. It gave us hands-on experience with OpenGL and helped us understand how a 3D game environment is designed and controlled.